Performance Evaluation

ITEC 4220 Adv Data Analytics
(Modified from CIS4930:
Data Science Foundations

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Performance Evaluation

Personal Loan Offer

- **Goal**: Identify customers who are more likely to accept the loan offer in future mailings
- **Outcome variable**: Accept loan (0/1)
- **Predictors:** Their last response to loan campaign, demographic info (Age, income, etc.), customers' relationship with bank (mortgage, securities account, etc.)

Personal Loan Offer

Logit(Personal Loan = Yes)

= −12.619 − 0.0325 Age + 0.0342 Experience

+0.0588 Income + 0.6141 Family + 0.2405 CCAvg

+0.0010 Mortgage − 1.0262 Securities_Account + 3.6479 CD_Account

−0.6779 Online − 0.9560 Credit Card

+4.1922 Education_Graduate + 4.3417 Education_Advanced/Professional.

Evaluating Classification Performance

Confusion matrix

Predicted	Actual	
	Positive, 1	Negative, 0
Positive, 1	70	20
Negative, 0	30	80

Positive = accept loan
Negative = reject loan

1. How many samples are in the data?
2. How many actually accepted the loan?
3. How many samples rejected?
4. How many were predicted to accept?
5. How many were predicted to reject?

Evaluating Classification Performance

Confusion matrix

Predicted	Actual		
	Positive, 1	Negative, 0	
Positive, 1	70	20	90
Negative, 0	30	80	110
	100	100	

Positive = accept loan
Negative = reject loan

1. How many samples are in the data? 200
2. How many actually accepted the loan? 100
3. How many samples rejected? 100
4. How many were predicted to accept? 90
5. How many were predicted to reject? 110

Evaluating Classification Performance

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Confusion matrix

Predicted	Actual	
	Positive, 1	Negative, 0
Positive, 1	70	20
Negative, 0	30	80

Positive = accept loan
Negative = reject loan

1. What % of samples were correctly classified?
2. What % of predicted “accept” cases were correct?
3. What % of actual “accept” cases were correctly identified?
4. What % of actual “reject” cases were correctly identified?

PATH FOR GEORGIA

Evaluating Classification Performance

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Confusion matrix

Predicted	Actual	
	Positive, 1	Negative, 0
Positive, 1	70	20
Negative, 0	30	80

Positive = accept loan
Negative = reject loan

1. What % of samples were correctly classified (accuracy)? (70+80)/200 = 75%
2. What % of predicted **"accept"** cases were correct (precision)? 70 / (70+20) = 78%
3. What % of actual **"accept"** cases were correctly identified (sensitivity)? 70 / (70+30) = 70%
4. What % of actual **"reject"** cases were correctly identified (specificity)? 80 / (80+20) = 80%

PATH FOR GEORGIA

Evaluating Classification Performance

- Confusion matrix

Actual	Predicted	
	Positive, 1	Negative, 0
Positive, 1	True Pos. (TP)	False Neg. (FN)
Negative, 0	False Pos. (FP)	True Neg. (TN)

- All = TP + FP + FN + TN
- Some common metrics:
  - ✓ **Accuracy** = (TP + TN) / All
  - ✓ **Precision** = TP / (TP + FP)
  - ✓ **Recall (Sensitivity)** = TP / (TP + FN)
  - ✓ **Specificity** = TN / (TN + FP)

Evaluating Classification Performance

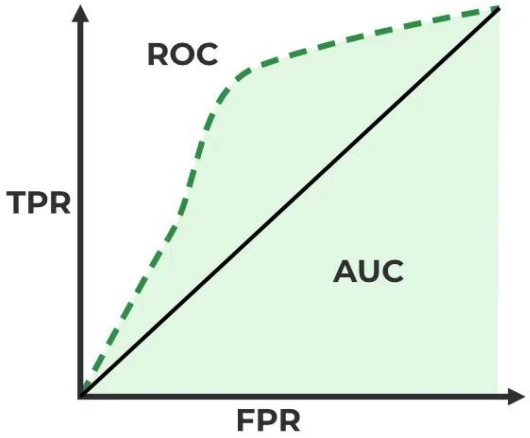
Confusion Matrix (Accuracy 0.9595)

Actual	Prediction 0	Prediction 1
0	1791	16
1	65	128

$$Accuracy = \frac{TP + TN}{All} = \frac{1791 + 128}{2000} = 0.9595$$

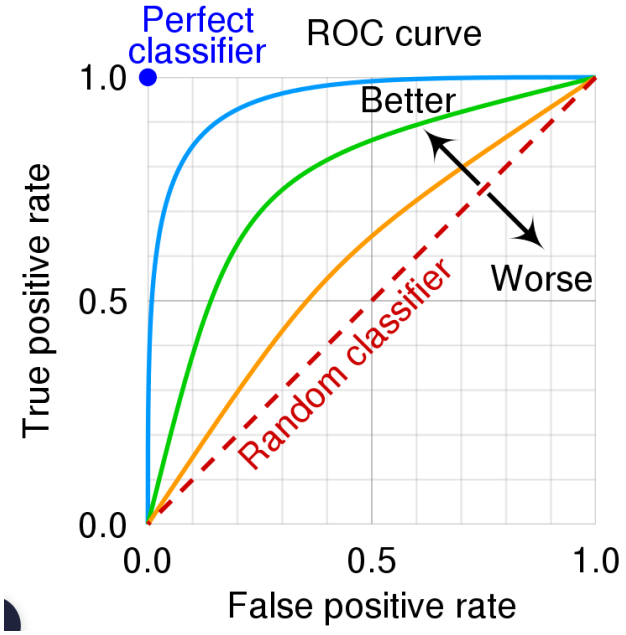
Evaluating Classification Performance

- ROC (Receiver Operating Characteristics)
  - ✓Plots the true positive rate (TPR) vs. the false positive rate (FPR)
  - ✓TPR = $\frac{TP}{TP+FN}$
  - ✓FPR = $\frac{FP}{FP+TN}$
- AUC (Area Under the Curve)



Evaluating Classification Performance

- A greater value of AUC shows a better model performance.
- Our main goal is to maximize this area to have the highest TPR and lowest FPR.



Additional Notes

- Additional considerations (see Chapter 5)
  - ✓ Maximize classification accuracy
  - ✓ Maximize sensitivity (When **false negatives** are costly or need to be minimized.)
  - ✓ Maximize precision or specificity (When **false positives** are costly or need to be minimized.)